Sophia Tsang

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Education

Cornell University, College of Engineering **Expected May 2027**
Bachelor of Science in Computer Science, Minor in Robotics **GPA: 3.923**
Relevant Courses: Systems Programming, Formal Verification, Frontiers of Computer Vision, Intro to ML, Foundations of Robotics, Robot Learning, Robot Perception, Intro to Game Architecture, Analysis of Algorithms, Differential Equations

Experience

MIT Lincoln Laboratory — Lexington, MA

Software Engineering and Robotics Intern | Python, PyTorch, TensorFlow, Docker, ROS2, VIO **May 2026 – Aug 2026**

- Engineered an end-to-end 3D terrain reconstruction pipeline for UAV imagery, adapting *Earth Observation Gaussian Splatting* and shadow mapping to generate depth maps, achieving **92.0% geometric accuracy** against LiDAR ground-truth terrain
- Integrated camera-derived depth and terrain maps into the3D scene-graph codebase and evaluation framework, enabling an alternative, physically grounded depth-estimation method that works with limited aerial perspectives for terrain reconstruction
- Architected asynchronous ROS2 processing with multiprocessing to decouple camera-stream ingestion from background depth estimation, enabling *non-blocking data processing and continuous depth-map generation*

Cornell Electric Vehicles — Ithaca, NY

Perception Lead, Autonomy Engineer | Rust, Python, C++, PyTorch, ROS2, Docker, Cargo **Oct 2024 – Present**

- Lead a 5-person Perception subsystem responsible for environmental sensing and world representation, overseeing feature development, project ownership, engineering milestones, and cross-team integration for vehicle perception systems
- Engineered *C++/Rust LiDAR driver*, owning the full data-processsing pipeline from UDP packet ingestion and IMU-based ego-motion compensation to *world-aligned 3D point cloud generation*, **reducing static-obstacle drift** from **0.92m to 0.04m**
- Optimized *two-layer-graph clustering* on **80k+ points** using azimuth-binning and range-adaptive thresholds, reducing scan latency to **20–30ms** and far-range over-merging from **99% to <10%** for *real-time obstacle detection and mapping*
- Developed *occlusion-tolerant multi-object tracking* and point-cloud-to-*occupancy-grid* generation, maintaing persistent obstacle identities and environmental state for *downstream planning and decision-making*

Noah Snavely Research Group — Roosevelt Island, NY

Research Intern | PyTorch, COLMAP, CUDA, Gemini, GCP, Flask, Render, React, Three.js, WebGL **June 2025 – Jan 2026**

- Engineered an automated geospatial data-collection pipeline integrating Google Maps API to identify landmark boundaries, interpolate candidate street-level viewpoints, and assemble georeferences imagery datasets across **4000+ landmarks**
- Developed a *geo-referencing pipeline* to recover absolute metric scale from SfM reconstructions using geographically tagged imagery and optimized viewpoint selection, co-authoring the ECCV 2026 paper, "Honey, I Shrunk the Arc de Triomphe"

Projects

Kernel::BSOD: OS Simulator | Rust, WebAssembly, JavaScript, React, Three.js, Cargo **Aug 2026 – Present**

- Architected a Rust-based kernel model compiled to WebAssembly, simulating a browser-hosted OS process management system through process control blocks, lifecycle state machines, and zombie-process cleanup to manage application instances
- Implemented a binary space partitioning (BSP) window manager in Rust, integrating process lifecycle events with UI rendering to dynamically allocate window geometry and reposition tiled windows based on user-triggered focus

Re:Color | Java, LibGDX, Agile Project Management **Jan 2026 – May 2026**

- Designed a color-based gameplay system with 7 other developers, enabling players to recolor the world by painting and absorbing color from the 2D platformer environment, and shipped the game *live* at GDIAC Showcase to **400+ attendees**
- Developed and shipped core player features and systems, including behavior-tree-driven player state management, animation synchronization, movement physics refinement, and a custom physics-based paint splatter rendering engine

Skills

Languages: Python, Rust, C/C++, C#, Java, Rocq, OCaml, JavaScript/TypeScript, SQL, HTML, CSS, Verilog
Frameworks/Libraries: PyTorch, NumPy, React.js, Three.js, WebAssembly, Flask, OpenCV, sqlite3, WebGL, Matplotlib
Tools: Git, CI/CD, Cargo, ROS2, Linux, Docker, VSCode, LibGDX, Unity, Render, GCP, Figma, Fusion 360, Slurm